

1. **2.83** The probability that a vehicle entering the Luray Caverns has Canadian license plates is 0.12; the probability that it is a camper is 0.28; and the probability that it is a camper with Canadian license plates is 0.09. What is the probability that
 - (a) a camper entering the Luray Caverns has Canadian license plates?
 - (b) a vehicle with Canadian license plates entering the Luray Caverns is a camper?
 - (c) a vehicle entering the Luray Caverns does not have Canadian plates or is not a camper?

2. **2.84** The probability that you are home when a telemarketing representative calls is 0.4. Given that you are home, the probability that goods will be bought from the company is 0.3. Find the probability that you are home and goods are bought from the company.

3. **2.85** The probability that a doctor correctly diagnoses a particular illness is 0.8. Given that the doctor makes an incorrect diagnosis, the probability that the patient files a lawsuit is 0.9. What is the probability that the doctor makes an incorrect diagnosis and the patient sues?

4. **2.88** Before the distribution of certain statistical software, every fourth compact disk (CD) is tested for accuracy. The testing process consists of running four independent programs and checking the results. The failure rates for the four testing programs are, respectively, 0.01, 0.03, 0.02, and 0.01.
 - (a) What is the probability that a CD was tested and failed any test?
 - (b) Given that a CD was tested, what is the probability that it failed program 2 or 3?
 - (c) In a sample of 100, how many CDs would you expect to be rejected?
 - (d) Given that a CD was defective, what is the probability that it was tested?

5. **2.99** Suppose that the four inspectors at a chocolate factory are supposed to stamp the expiration date on each package of chocolate at the end of the assembly line. John, who stamps 20% of the packages, fails to stamp the expiration date once in every 200 packages; Tom, who stamps 60% of the packages, fails to stamp the expiration date once in every 100 packages; Jeff, who stamps 15% of the packages, fails to stamp the expiration date once in every 90 packages; and Pat, who stamps 5% of the packages, fails to stamp the expiration date once in every 200 packages. If a customer complains that her package of chocolate does not show the expiration date, what is the probability that it was inspected by John?